

1. Runtime

Definition: Runtime is the phase when a program is actually running and executing instructions on the computer.

Example: User input is read and processed while the program is running.

2. Compiletime

Definition: Compiletime is the phase when source code is translated into machine code before the program runs.

Example: Syntax errors like a missing semicolon are caught during compilation.

3. OOP (Object-Oriented Programming)

Definition: OOP is a programming approach that organizes code into objects that contain data and behavior together.

Example: A `car` object has data like `speed` and functions like `accelerate()`.

4. Encapsulation

Definition: Encapsulation is hiding internal data and allowing access only through controlled methods.

Example: Making a variable `private` and accessing it using `get()` and `set()` functions.

5. Abstraction

Definition: Abstraction shows only essential features and hides unnecessary implementation details.

Example: Using a `drive()` function without knowing how the engine works internally.

6. Inheritance

Definition: Inheritance allows one class to reuse and extend the properties of another class.

Example: A `Developer` class inheriting common fields from an `Employee` class.

7. Polymorphism

Definition: Polymorphism means one interface can represent different behaviors.

Example: The same function name `draw()` behaving differently for `Circle` and `Square`.

8. Runtime Polymorphism

Definition: Runtime polymorphism is method selection decided during program execution using base class pointers.

Example: A base pointer calling a derived class's overridden function.

9. Compiletime Polymorphism

Definition: Compiletime polymorphism is method selection decided at compile time

using function overloading or operators.

Example: Multiple `add()` functions with different parameter types.

10. Use of `virtual` keyword in C++

Definition: The `virtual` keyword ensures the correct overridden function is called based on the object type at runtime.

Example: Marking a base class function as `virtual` so the derived version runs when accessed via a base pointer.